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Quantitative Macro-Credit Risk Modeling

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Abstract

Most traditional credit risk models are primarily based on logistic regressions with the assumption that the relationship between the borrower's characteristics and the probability of default (PD) remains static over time. Past systemic shocks, such as the 2008 financial crisis and the COVID-19 pandemic, have shown that such an assumption does not hold because these models often fail to capture macroeconomic shifts, resulting in predictive instability. This project proposes a regime-dependent credit risk model that captures conditioned probability of default (PD) on three frontiers: individual borrower features, real-time macroeconomic indicators, and latent market regimes. The project uses high frequency dataset from Lending Club loan records merged with FRED macroeconomic data series. The project employs Hidden Markov Model (HMM) and Gaussian Mixture Models (GMM) for economic state detection, Extreme Gradient Boosting (XGBoost) for predicting PD given borrower features, macroeconomic states, and latent market regimes. Results indicate that the macro-augmented (Model 2) and the regime-dependent (Model 3) models perform better with ROC-AUC of 0.705 and 0.715, compared to 0.683 for the base model (uses borrower only features). Additionally, the models have accuracy scores of 0.632, 0.968, and 0.973 for models 1, 2, and 3, respectively. This result suggests that incorporating macroeconomic conditions and market regimes improves the prediction of default rates.

Keywords: Credit Risk, Machine Learning, XGBoost, Hidden Markov Models, Macroeconomic Conditioning, Regime-Switching, Probability of Default.

1. Introduction

Credit risk models have long relied on logistic regression and scorecard-based approaches. These methods work reasonably well under normal conditions, but they share a fundamental assumption that tends to break down in practice: that the link between a borrower's characteristics and their likelihood of defaulting stays roughly constant over time. The 2008 financial crisis and the COVID-19 shock made this assumption hard to defend. Default rates spiked in ways that static models simply didn't anticipate, not because the borrowers looked different on paper, but because the macro environment had shifted entirely. A model trained on 2005-2007 data had no way of knowing what a 10% unemployment rate or near-zero interest rates would do to repayment behavior.

This project investigates the effect of the shift in macroeconomic conditions on loan repayment behavior. Instead of treating credit risk as a fixed function of borrower features, we want to condition it on the macroeconomic environment and on which "regime" the economy is in at a given point in time, whether that's a period of expansion, stress, or something in between.

The Fama-French model gave us a useful way to think about this. Just like equity returns can't be explained by market beta alone and benefit from adding size and value factors, credit risk probably can't be captured by borrower characteristics alone. There's a systematic component driven by macro conditions and market regimes that static models ignore.

This research uses models that treat probability of default (PD) as a function of: the borrower's own features, macroeconomic variables at the time of assessment, and a latent regime label estimated from the data. The core question we want to answer is whether adding those last two components actually improves prediction in a meaningful way, not just on paper, but across different economic periods, including stress scenarios.

2. Research Questions and Objectives

This project follows the following key research questions:

1. Does incorporating macro-economic conditions in credit risk modelling improve the predictive performance of the probability of default of macro-economic conditions + base (borrower only) model versus the traditional borrower-only model?
2. Does the incorporation of latent economic regimes in credit risk modelling improve the performance of the model further, especially in times of economic turbulence?
3. Which regime shift detection methods best capture meaningful economic states for modelling in credit risk?
4. Do regime-aware models contribute to improvements in credit risk model calibration and stability of probability of default estimates across different economic cycles?
5. Does the proposed framework, i.e., regime-dependent stress-testing framework, generalize well in different economic cycles (calm, turbulent, recession, etc)

The project aims to address these research questions by executing the following objectives:

1. Integrating and synchronizing multiple data sources, including the integration of the Lending Club loan dataset with FRED macroeconomic data series (e.g., Unemployment Rate). We achieve this using a point-in-time join method, ensuring no data leakage.
2. Implementing and comparing the Hidden Markov Model, Gaussian Mixture Models, and K-means clustering, and detecting the most statistically robust economic regimes.
3. Iteratively developing a three-tier modelling hierarchy using the Gradient Boosting (XGBoost) algorithm:

   Model 1: Borrower’s characteristics only (Traditional Model)

   Model 2: Borrower’s characteristics + macroeconomic features

   Model 3: Borrower + Macro + Regime-label features

4. Comprehensively assessing the performance of the three models using AUC and calibration curves to determine if the regime-awareness of a model decreases probability drift during periods of economic shift.
5. Performing stress testing simulation of extreme economic shocks such as the 2008 economic crisis and Covid-19, and evaluating the stability of probability of default estimates, and generalizing the results across different market conditions.

3.0 Literature Review

The 2008 financial crisis pushed credit risk modeling to the top of the research agenda in ways it had not been before. Banks lost enormous amounts of money, and a big part of the reason was that their models were not built to handle the kind of conditions that emerged. Research has moved steadily toward approaches that can handle non-linear relationships and adapt to shifting economic conditions, rather than assuming the world stays roughly the same year after year.

This review covers three areas that directly shape this project. First, the classical credit risk models that still dominate practice, and where they consistently fall short. Second, the growing body of work on connecting default probability to the broader macroeconomic environment. Third, the regime-switching literature, which addresses the fact that borrower behavior and default patterns do not look the same across all economic periods.

3.1 Traditional Credit Risk Models and Their Limits

Most discussions of credit risk in the academic literature start with Merton (1974). His structural model frames default as the point where a company's asset value drops below what it owes, a clean piece of theory, but one that depends on inputs that cannot be directly observed in practice. That gap between theory and application pushed researchers toward reduced-form models, notably Jarrow and Turnbull (1995) and Duffie and Singleton (1999), which treat default as a probabilistic event governed by a hazard rate rather than something derived from unobservable asset dynamics. These caught on in practice but came with their own limitations: they still assume the relationship between a borrower's characteristics and their default probability does not change much over time.

The logistic regression model, which predicts the probability of a binary outcome (i.e., yes/no, 0/1, etc), has been the default tool in consumer lending for decades. Altman's Z-score (1968) is the most cited version of this idea, a fixed weighted combination of financial ratios used to classify firms as distressed or not. The model has been updated and extended many times, but the underlying structure has stayed the same: a linear mapping from inputs to risk, estimated once and applied going forward. Bello (2023) compared these classical methods to more recent machine learning alternatives and found that logistic regression holds up well when data is limited, but gets overtaken when non-linear patterns are present. Suhadolnik et al. (2023) frame the problem more bluntly: the real issue is not that the model is too simple, but that it treats economic conditions as irrelevant, which they argue is a structural flaw rather than a tuning problem.

3.2 Machine Learning Applied to Credit Risk

Ensemble methods started getting serious attention in credit risk around 2015. Ensemble methods are machine learning techniques that combine several individual models (known as base models or learners) via boosting, bagging, or stacking to produce a single, more accurate and robust model. Lessmann et al. (2015) compared 41 classifiers across multiple datasets and found that boosting methods consistently deliver stronger Area Under the Receiver Operating Characteristic Curve (ROC-AUC) scores than classical alternatives. They were careful to flag something important, that is, a strong ROC-AUC does not guarantee well-calibrated probabilities, and in credit risk, that distinction matters. A lender needs PD estimates that reflect actual default rates, not just a model that ranks borrowers correctly relative to each other. Brown and Mues (2012) made a complementary point about gradient boosting specifically: it handles missing values and feature interactions better than logistic regression and does not require the extensive manual feature engineering that traditional scorecards demand.

One pattern worth noting across this literature is how often the time dimension gets ignored in model evaluation. Random train-test splits are still common practice in published studies, which can produce results that look good on paper but would not hold up in deployment. A model trained on data from 2015 and evaluated on data from 2010 has effectively been given information it could not have had at the time. This project avoids that by using a strict time-based split throughout, with evaluation only on loans issued after a fixed cutoff date.

3.3 Macroeconomic Conditioning in Credit Models

Connecting credit risk to macroeconomic conditions is not a new idea, but embedding it properly into a machine learning pipeline is still relatively underdeveloped. Wilson's CreditPortfolioView (1997) was one of the earlier frameworks to do this systematically, modeling portfolio-level default rates as a function of GDP growth, unemployment, and interest rates through a system of equations. It showed that macro variables carry real explanatory power for default behavior at the aggregate level, even if firm-level characteristics dominate at the individual borrower level.

The regulatory framework reflects this too. Basel II and III draw a formal distinction between through-the-cycle and point-in-time PD estimates (BCBS, 2006). Through-the-cycle estimates try to average out cyclical variation; point-in-time estimates reflect where risk stands given current economic conditions. A model that conditions on macro variables is essentially doing the latter; it is trying to capture how risk shifts as the economy moves through different phases, rather than treating the cycle as noise to be smoothed away.

The choice of macro variables in this project is grounded in empirical findings. Figlewski et al. (2012) found that unemployment and industrial production growth have statistically significant effects on corporate default rates, holding firm-level factors constant. Blanco et al. (2020) showed that high-yield credit spreads, including the BAMLH0A0HYM2 series, carry forward-looking information about aggregate default risk that borrower data alone does not reflect. Both findings point toward the same conclusion: leaving macro variables out of a credit model means leaving out information that genuinely moves default rates.

3.4 Regime Switching Framework

Hamilton (1989) introduced Hidden Markov Models to economics through a deceptively simple observation: US GDP growth does not behave as if it were drawn from a single stable distribution. It behaves more like a process that switches between two states, expansion and contraction, with the active state hidden from direct observation but inferable from the data. That idea opened up a whole line of research into latent regime models for financial and economic time series.

The implications for credit risk are significant. Nickell et al. (2000) showed that credit rating transition probabilities vary substantially across economic states. A borrower sitting at a given rating is meaningfully more likely to be downgraded during a recession than during an expansion, even if their underlying financials look similar. A model that assumes these probabilities are constant is going to get things systematically wrong every time the economic environment shifts.

Bai et al. (2019) brought this logic into a machine learning context by training separate gradient boosting models for each regime identified by a two-state Hidden Markov Model

(HMM). Their regime-specific models outperformed a single pooled model on both ROC-AUC and Brier score, which is probably the most direct empirical precedent for what this project does. The approach here extends theirs in a few ways: three regimes instead of two, multiple clustering methods compared rather than HMM alone, and the regime label included as a feature in the model rather than used to partition the training data entirely, which keeps the model from losing information during regime transitions.

Gaussian Mixture Models are also included as an alternative to HMMs. Frühwirth-Schnatter (2006) makes the case that GMMs can be more appropriate when there is no strong reason to assume regime transitions are Markovian. Whether that assumption holds for the specific macro feature set used here is an open empirical question, and comparing the two methods directly is part of how this project tries to answer it.

4.0 Methodology & Approach

We propose a regime-dependent credit risk model that incorporates borrower features, macroeconomic conditions at a given time t, and latent market regimes to evaluate the creditworthiness of a borrower by assessing the probability of default of the borrower. We implemented a multi-step quantitative approach, moving from synthesis of raw data to unsupervised regime detection (via Hidden Markov Model), and finally to supervised predictive modeling (via Extreme Gradient Boosting). The guiding equation is:

$$Probability\ of\ Default\ (PD) = P_r(Default|X_{borrower},\ M_t,\ R_t) \quad \textbf{.........(1)}$$

Where:
$X_{borrower}$: Borrower-specific characteristics
M_t: Macroeconomic variables at time t
R_t: Latent market regime at time t

4.1 Data Synthesis

Our data layer comprises two primary data sources:

Micro-level data (Loan Lending data), which we obtained from Lending Club, represent comprehensive peer-to-peer loan records. This dataset was the primary record for borrower features, including FICO score, Debt-to-Income Ratio, employment length, and loan status, i.e., default vs. non-default. This data set formed the basis for understanding the behavior of a borrower, which is key when analysing the credit risk and making decisions on whether to approve or decline a loan request from borrowers. **Macro-level data**, high-frequency data, were obtained from FRED. This data captures the economic conditions, such as market stress (Credit Spread, proxy for market stress), systemic risk, consumer health (proxied by unemployment rates, consumer price index, etc), industrial production index, interest rates (proxied by federal funds effective rates), and real gross domestic product. We incorporated borrower-level data with macroeconomic data that captures real economic activity, monetary conditions, and market stress to integrate the systemic economic effect in credit risk modeling. The variables are selected on the basis of established empirical evidence. All macroeconomic data were obtained from Federal Reserve Economic Data (FRED), ensuring consistency, transparency, and reproducibility. The dataset comprised variables with mixed frequencies, which were systematically transformed and aligned to a monthly frequency, matching the temporal structure of loan issuance data from Lending Club Loans Data.

We implement a point-in-time join to merge micro and macro-level data i.e., macro-indicators with loan records on the basis of loan issuance time. This ensures that the model only makes use of economic data available at the time of credit evaluation to avoid data leakage. The equation for Xthis is shown below:

$$D_{merged} = \{(x_{i,t}, m_{\tau}) | \tau = max(t_{macro} \leq t_{loan})\} \text{ ……….. } \mathbf{(2a)}$$

Where:
$x_{i,t}$: features for the borrower i at the loan issuance time t
m_{τ}: microeconomic indicator at time τ
t_{loan}: time stamp of credit decision

All the macroeconomic variables were lagged by one period to prevent look-ahead bias. Macroeconomic information available at the time $t - 1$ is used for each loan issued at a time t:

$$X_t^{macro} = X_{t-1}^{macro} \text{ ……………….. } \mathbf{(2b)}$$

The lag structure (equation 2b) ensures that the model does not include information that would not be observable at the time of credit scoring.

4.2 Regime Detection via Hidden Markov Model, HMM

We implemented the Hidden Markov Model (HMM) with Gaussian emission to help identify latent economic states, before training our regime-aware predictive model. The Hidden Markov Model was trained on macroeconomic indicators with a monthly frequency, and the expected output was discrete regime labels, **R_t = {0, 1},** which represent expansion and stress states of the economy for every period in the data set.

The equation below gives the probability of observing a sequence of macroeconomic indicators, O, given latent economic regimes, Q:

$$P(O|\lambda) = \Sigma_Q \pi_{q1} b_{q1}(O_1) \prod_{t=2}^{T} a_{qt-1} b_{qt}(O_t) \text{ …………. } \mathbf{(3)}$$

Where:
$a_{ij} = P(q_t = S_j | q_{t-1} = S_i)$: Transition probability between states
$b_j(O_t) \sim N(\mu_j, \Sigma_j)$: Gaussian emission probability for the states j
$S_t \epsilon \{0, 1\}$: Discrete regime labels assigned to period t

The model parameters $\theta = \{\pi, \mu_j, \Sigma_j\}$ were estimated via maximum likelihood estimation (MLE) using the expectation maximization algorithm

4.3 Supervised Modeling Framework

We utilized the Extreme Gradient Boosting model primarily due to its effectiveness in handling non-linear interactions and missing data more efficiently. We iteratively modelled credit risk starting with a baseline model, trained on specific borrower characteristics, X. The XGBoost loss function is defined as:

$$\mathcal{L}(\phi) = \sum_i l(\hat{y_i}, y_i) + \sum_k \Omega(f_k) \text{ ……………………… } \mathbf{(4)}$$

Thus, the baseline model is defined by the equation:

$$PD = \hat{y} = f(X_{borrower}) = Pr\left(default \mid X_{borrower}\right) \text{......... } \mathbf{(5)}$$

We then geared up to the Macro-augmented model, which combines both the borrower characteristics, $X_{borrower\ features}$, and macro-economic conditions m_t. The equation for this model is defined as:

$$PD = \hat{y} = f\left(X_{borrower}, m_t\right) = Pr\left(default \mid X_{borrower}, m_t\right) \text{........... } \mathbf{(6)}$$

Finally, we incorporated latent regime in our regime-dependent credit risk model, allowing the model to learn split point for different regimes, borrower features, and economic state. This model is defined by the equation:

$$PD = \hat{y} = f\left(X_{borrower}, m_t, R_t\right) = Pr\left(default \mid X_{borrower}, m_t, R_t\right) \text{......... } \mathbf{(7)}$$

The output is a probability measure of the likelihood of a borrower defaulting on their loan obligation, conditioned on their past behavior, past behavior and macroeconomic conditions, and past behavior + macroeconomic conditions + market regimes.

4.4 Model Evaluation and Stress Testing

We performed model evaluation using the precision-recall AUC (PR-AUC), accuracy, and ROC-AUC metrics as the primary model evaluation frameworks. We used the Brier Score and calibration curves to evaluate the model calibration of probability of default estimates. We performed out-of-sample stress testing by conducting a crisis simulation, which was achieved by shifting the macro-input features manually to the 2008 financial crisis level and then observing the sensitivity and migration of probability of default scores across the models

Figure 1 below provides a visual of the workflow of our project from developing data pipeline to stress testing and scenario analysis.

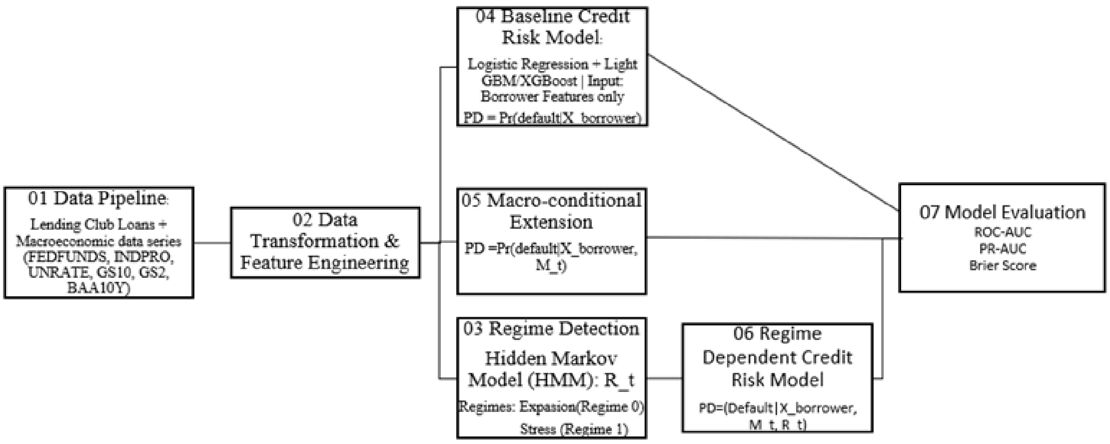

Figure 1: Project workflow

5.0 Results & Discussion

5.1 Hidden Markov Model Regime Identification

The Hidden Markov Model (HMM) identified two latent macro-financial regimes that persisted for the selected sample period spanning 2008 – 2018. We utilized transformed macroeconomic features, specifically, the lagged version of unemployment, yield curve, credit spread, consumer price index, and industrial production due to their strong cyclical and financial stress signaling properties. A two-state Gaussian HMM with diagonal covariance structure and smoothed macroeconomic input features produced the most economically stable and interpretable regime classification (See **Figures 2a & 2b**).

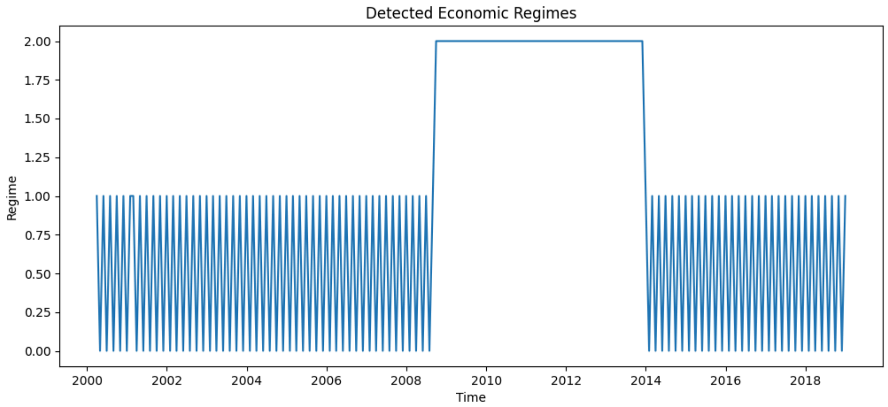

Figure 2a: Initial HMM results exhibited excessive regime switching due to high-frequency noise in macro-financial variables. This was mitigated through feature selection and smoothing, and reducing n_components from 3 to 2, resulting in economically interpretable regimes (**Figure 2b)**.

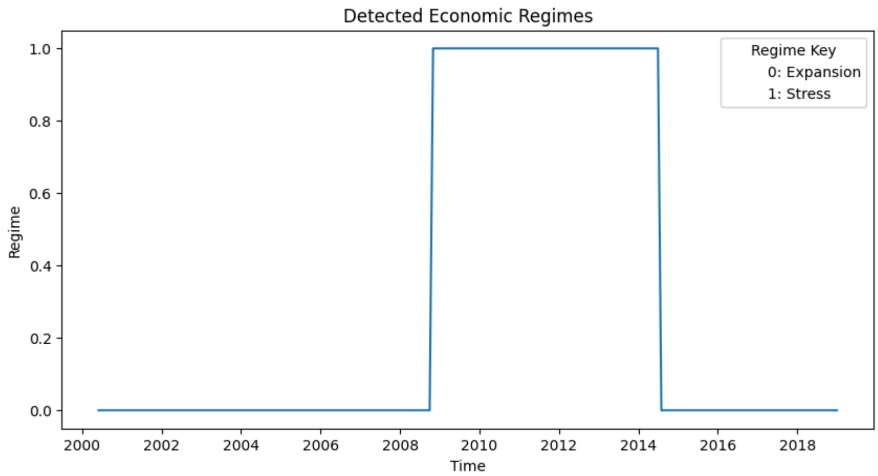

Figure 2b: Detected regimes

As shown in **Figure 2b**, the resulting regime dynamics indicated prolonged periods of macroeconomic stability, which are interrupted by a sustained interval of elevated macroeconomic and financial stress. HMM classified the first regime as 'Normal/Expansion Regime,' which was characterized by relatively low unemployment, steeper yield curve, and lower credit spread, indicating accommodative credit conditions and stronger economic activity (**See Table 1**). The second regime detected was 'Stress/Recovery Regime,' which exhibited persistently heightened unemployment levels, widened credit spread, flatter yield curve, and reduced industrial production growth, consistent with poor credit environments and stricter financial conditions (See **Table 1**).

Notice that the detected stress regime is broadly aligned with the aftermath of the Great Financial Crisis. The HMM captured the extended post-crisis adjustment period, spanning from around 2010 to 2015, which was characterized by increased systemic risk and prolonged macroeconomic weakness. This observation is economically intuitive because during this period, labor market deterioration, impaired credit intermediation, and increased risk aversion persisted for several years after the crisis.

Table 1: Regime vs Macroeconomic Variables

Features	Regime 0 (Expansion)	Regime 1 (Stress)
Unemployment_Lag1	4.9910	8.3773
Consumer_Price_index_inflation_Lag1	0.0020	0.0012
Industrial_Production_Growth_Lag1	0.0006	0.0004
Credit_Spread_Lag1	2.4070	3.1775
Yield_Curve_Lag1	1.0688	2.1331

Table 2: Regime Transition Matrix

	Regime 0 (Expansion)	Regime 1 (Stress)
Regime 0 (Expansion)	0.9935	0.0065
Regime 1 (Stress)	0.0145	0.985

The regime transition matrix (See **Table 2**) further demonstrates high regime persistence, showing that the identified latent states were not driven by short-term noise or transient fluctuations. The persistence of the identified regimes is consistent with economic intuition that is macroeconomic conditions change gradually over time. Additionally, regime classification exhibited strong correspondence with movements in corporate credit spreads. Notably, periods classified as stress periods coincided with increased levels of credit spread widening, suggesting that the HMM effectively captured latent systemic risk conditions relevant to credit default behavior. This observation provides economic justification for the use of regime information in credit risk modeling.

5.2 Performance of Nested XGBoost Models

Three XGBoost models were developed to evaluate whether the addition of macroeconomic variables and latent regime information increases the predictive value in credit default prediction. The first model (base model) used borrower features only, including loan amount, debt-to-income ratio, interest rate, installment obligation, FICO score, revolving utilization, and annual income as model input features to predict the probability of default (see **Equation 5**). The second model incorporated lagged macroeconomic variables in addition to borrower features (see **Equation 6**), while the third model further incorporated latent regimes estimated from the HMM framework (see **Equation 7**).

Predictive results obtained indicated consistent and monotonic improvements in performance across all evaluation metrics used as macroeconomic information was added to the modeling framework.

Table 3: Summary of Model Performance

Model & Features Included	ROC-AUC	PR-AUC	Brier Score
Model 1: Borrower Features	0.6830	0.0268	0.2179
Model 2: Borrower + Macroeconomic Variables	0.7050	0.0382	0.0632
Model 3: Borrower + Macroeconomic Variables + Regime State	0.7150	0.0421	0.0575

As can be observed from **Table 3**, the base model (borrower features only) achieved moderate discriminatory performance with ROC-AUC of 0.6830, PR_AUC of 0.0268, and Brier Score of 0.2179, suggesting that characteristics of a borrower contain meaningful but incomplete information regarding credit default risk. The comparatively poor calibration (see **Figure 4**) and low precision-recall score indicate that borrower features alone are not enough to accurately estimate probabilities of default in evolving economic environments. Incorporation of macroeconomic variables in Model 2, resulted in substantial improvements in both the discriminatory and calibration metrics. Model 2 achieved ROC-AUC of 0.7050, PR_AUC of 0.0382, and Brier Score of 0.0632. Compared to Model 1, Model 2 achieved ROC-AUC and PR-AUC margin increments of 2.2% and 1.14%, respectively, suggesting that macroeconomic conditions contribute materially to rates of credit defaults. It appears that deteriorating labor market conditions, widening credit spreads, and adverse term structure movements influence borrower loan repayment ability beyond individual borrower characteristics.

Model 3 achieved the strongest performance, which included latent regime information from HMM in addition to borrower features and macroeconomic variables. The ROC-AUC, PR-AUC, and Brier Scores for Model 3 were 0.7150, 0.0421, and 0.0575, respectively. Although the incremental ROC-AUC and PR-AUC increased relative to Model 2, the inclusion of regime information produced superior rare-event detection capability and the lowest Brier score in all three models. This observation suggests that latent macro-financial regimes are able to capture non-linear systemic effects which may not be fully represented by observable macroeconomic variables alone. This fact explains the superior performance of Model 3 compared to Models 2 and 1.

5.3 Precision-Recall and Calibration Analysis

Analysis of precision-recall curves for the three models further reinforced the superiority of regime-aware credit risk modeling (Model 3). It can be observed from **Figure 3** that the regime-enhanced model consistently dominated the borrower features-only model across recall levels, indicating stronger rare-event detection performance in an imbalanced default prediction setup. This result is economically relevant because, as often expected, periods of elevated systemic stress are usually associated with clustering of defaults and non-linear deterioration in borrower creditworthiness. Credit risk models incorporating borrower features alone do not provide accurate probabilities of default because they fail to account for such macro-financial interactions. Regime-aware models, on the other hand, adapt dynamically with the changing economic conditions; they provide a more robust measure of the probabilities of default.

The calibration curve shown in **Figure 4** confirms similar observations and conclusions that have already been made. As can be observed from the calibration curve, borrower features only model (Model 1) exhibited substantial deviation from the ideal model calibration line, indicating systematic overestimation of the probabilities of default. In contrast, Model 2 and Model 3 curves show substantially improved calibration behavior with significantly lower Brier Score values of 0.0632 and 0.0575, respectively. This improved calibration performance indicates that macroeconomic variables and latent macro-financial conditions contain valuable information regarding the underlying default environment. This is key, especially in applications involving the estimation of expected loss, capital allocation, stress testing, and risk-sensitive pricing, where accurate estimation of probability is a key discriminatory power.

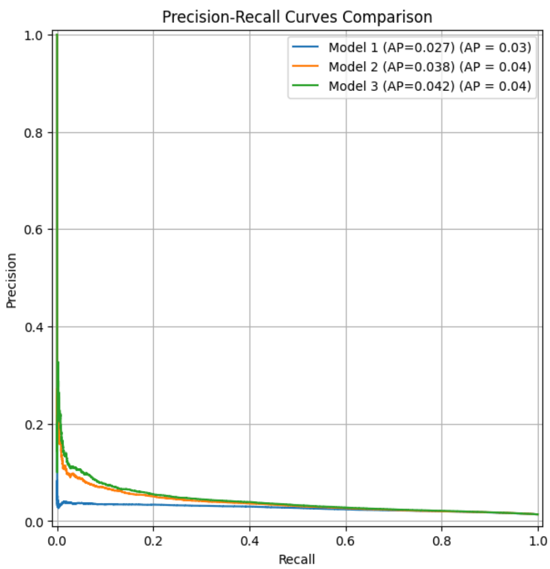

Figure 3: Precision-Recall Curve

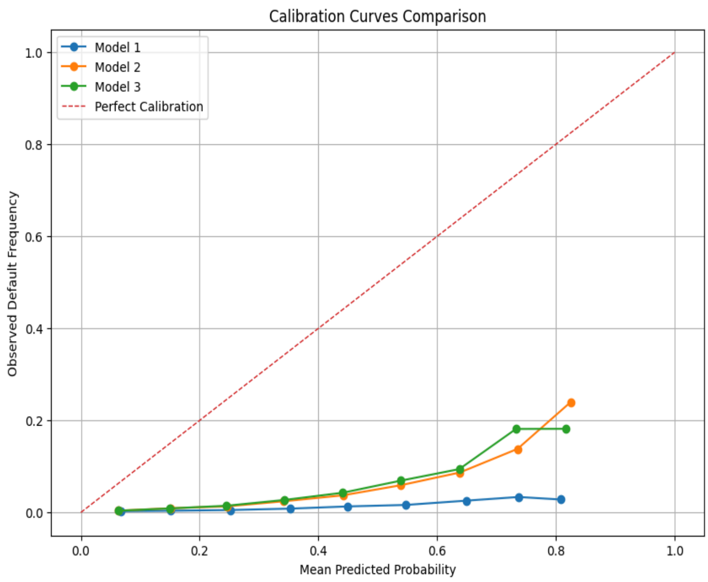

Figure 4: Calibration Curves

5.4 Economic Interpretation and Implication

Our empirical findings in this paper support the hypothesis that credit default risk is state-dependent and is impacted by broader macro-economic conditions, and that borrower-only features do not fully explain default dynamics, especially during periods of increased systemic stress. Latent economic regimes impact default dynamics via multiple transmission channels, including labor market deterioration, tight credit conditions, declining borrower liquidity, and stringent refinancing constraints. As indicated by the results, during a stressful economic regime, even individuals with stable borrower characteristics may experience increased probabilities of default as a result of macroeconomic spillovers. The observed model performance improvements for the regime-aware model further suggest that non-linear economic dynamics are critical for accurate credit risk estimation and, therefore, the application of traditional static credit scoring models underestimates systemic default risk during periods of stressful macroeconomic conditions.

The findings in the paper demonstrate the value of incorporating machine learning frameworks with macro-financial regime detection techniques in credit risk modeling. The results give strong evidence that macroeconomic conditions and systemic regimes contain incremental predictive information, justifying the use of regime-sensitive machine learning frameworks in modern credit risk management. As such, financial institutions can benefit from incorporating latent economic state information into probability of default estimation frameworks to improve model robustness, calibration quality, and stress period forecasting performance.

6.0 Conclusion

This study investigated the impact of macroeconomic conditions and latent economic regimes on credit default prediction using a machine learning framework combining Hidden Markov Model for regime detection and XGBoost for evaluating the probability of credit default. The primary goal was to integrate systematic macro-financial information and assess whether it improves the predictive power relative to traditional borrower-level credit risk models. The empirical findings indicate that macroeconomic variables and latent regime information provide substantive incremental predictive value beyond what the borrower-only model provides. While the borrower-only model (Model 1) achieved moderate discriminatory performance, the inclusion of macroeconomic variables significantly improved both the ROC-AUC and PR-AUC. Further integration of latent regime information produced further improvements across ROC-AUC, PR-AUC, and Brier Scores, indicating that hidden macro-financial states contain useful nonlinear information for credit default modeling.

The Hidden Markov Model successfully isolated economically meaningful latent regimes characterized by distinct macro-financial environments. The stress regime detected by HMM broadly aligned with periods of increased systemic risk, widened credit spreads, and weak labor market conditions typical of periods following a global economic crisis. The results further indicated that default risk is inherently state-dependent and strongly influenced by broader economic conditions; thus, borrower features alone are not sufficient to effectively model credit risk. The study demonstrates the effectiveness of incorporating probabilistic regime detection techniques with modern machine learning frameworks in credit risk modeling. The findings provide supportive evidence that macro-financial machine learning frameworks can improve model robustness and improve predictive stability, especially for evolving economic environments.

Finally, this study contributes to the growing literature on macro-financial credit risk modeling by showing that latent economic regimes contain useful predictive information for consumer loan default prediction and that regime-dependent machine learning models offer superior results over traditional static credit scoring approaches.

7.0 Future Direction

Future research work can build upon the findings of this study by substituting the Hidden Markov Model framework in regime detection with deep sequence models like Long Short-Term Memory (LSTM) to detect more complex, long-term temporal dependencies in macro-financial data. Also, future research should integrate Explainable Machine Learning techniques like Shapley Additive explanations (SHAP) to provide the necessary transparency as to how borrower features interact with macroeconomic data within the XGBoost model across different regime states. Finally, in order to establish structural generalizability, future research should expand the empirical validation of this macro-financial machine learning framework through the use of diverse asset classes, such as mortgages or corporate loans, within broader geographical markets.

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